

YOLOv7-Based Vehicle Detection for Adaptive Traffic Signal Control

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Abstract—One of the key problems in cities is traffic congestion, which leads to the expansion of travel time, fuel use, and pollution of the environment. This paper puts forward an AI-based smart traffic management system that will dynamically adjust the timing of the traffic lights according to the real-time traffic density that is computed by taking images from cameras at the points where the roads intersect. The suggested system is based on computer vision and deep learning methods, where a YOLOv7 object detection model is applied to the traffic images to accurately identify and classify vehicles. The number of vehicles that is identified in each lane is used in estimating the density of traffic and setting adaptive signal timing, whereby high traffic density has the highest green signal timing so as to minimize unwarranted wastage and maximize the efficiency of the flow. Moreover, there is an interface for traffic conditions and signal status visualization in real time, which allows effective monitoring of the traffic and manual intervention in cases of necessity. The system is tested on various traffic images under various conditions of congestion, and the results of experiments prove a significant decrease in the waiting time of vehicles and the general time of congestion in comparison with the standard fixed-time traffic signal control systems. The proposed solution is implemented using Python and provides a cost-efficient and scalable way of working with intelligent transportation systems and smart city traffic control.

Index Terms—Artificial Intelligence, Computer Vision, Object Detection, Traffic Signal Control, Intelligent Transportation System, OpenCV, Deep Learning.

I. INTRODUCTION

In the contemporary urban setting, traffic congestion is an issue of serious concern because of the fast growth in the number of cars and the comparatively small growth of road networks. Delay at intersections is prolonged, which leads to unnecessary travel delays, increased fuel consumption, environmental pollution, and lower traffic efficiency. Traditional traffic signal systems are mainly based on fixed-time control strategies, where the time taken by the signals is fixed in advance through historical traffic data. These systems may be easy to deploy, but they lack the ability to respond to real-time changes in traffic, thus resulting in poor signal utilization, unnecessary waiting time, and poor performance during peak and off-peak hours.

Some previous traffic control strategies tried to enhance traffic movement by using manual surveillance, loop detectors, infrared, or inductive sensors on road surfaces. Nonetheless, sensor-based systems are expensive to install and operate, as well as being subject to environmental failure. Addition-

ally, conventional image-processing-based techniques were based on hand-crafted features and threshold-based techniques, which are prone to variations in lighting, occlusions, and intricate traffic situations. Consequently, in real-world environments, these methods are subject to flawed vehicle detection and imprecise estimations of traffic density.

The latest developments in artificial intelligence and deep learning have allowed the development of more effective solutions for traffic analysis based on vision-based systems. Object detection models based on deep learning are more accurate and better generalizable than conventional methods. This project uses a deep learning framework that is built based on the YOLOv7 algorithm to identify and categorize vehicles in traffic at road intersections. The driving force behind the choice of a YOLOv7-based approach is its real-time detectability, high detection rate, and ability to handle dense and intricate traffic conditions. The system does not have the weaknesses of fixed-time regulation and traditional sensor-based systems, as it accurately estimates the number of vehicles in each lane.

The suggested smart traffic management system makes use of the results of vehicle detection to estimate traffic density and dynamically change the timing of traffic signals. Lanes that have a greater number of vehicles are assigned a longer green signal duration, whereas lanes that have fewer vehicles are assigned a shorter green signal duration. This adaptive plan guarantees improved use of the existing road capacity and ensures that a significant amount of unnecessary waiting at intersections is avoided.

There is also a Graphical User Interface (GUI) that provides a graphical representation of the number of vehicles, the state of the signals, and the movement of the traffic, which can be controlled automatically and monitored manually.

The primary goals of the research are as follows:

- To develop a framework based on AI that can be used to identify and process traffic around real-time camera images.
- To implement an adaptive signal-control algorithm that adjusts timing based on real-time density estimation.
- To create an easy-to-use interface to visualize and monitor traffic conditions.

The rest of this paper is structured in the following way. Section II outlines the corresponding work and the current methods of managing traffic and their weaknesses. Section

III outlines the methodology and system architecture of the proposed smart traffic management system, including vehicle detection and adaptive signal control. Section IV presents the experimental results and discussion, where the effectiveness of the offered system and its comparison with traditional methods can be observed. Section V is the final part of the paper, where possible future work directions are outlined.

II. RELATED WORK

The traditional urban traffic management measures mostly depend on fixed-time and actuated signal control methods formulated with old traffic data records. However, these systems are not sufficient in the case of modern urban settings, which experience highly dynamic traffic flows due to peak hours, accidents, weather conditions, and road operations. Such restrictions normally lead to overcrowding, longer queues, more fuel wastage, as well as poor use of the roads. Therefore, recent studies have drawn attention to smart traffic management strategies utilizing the opportunities of artificial intelligence, machine learning, and computer vision to allow real-time traffic monitoring, dynamic signal regulation, and less infrastructure reliance.

Sridevi et al. [1] This research was able to indicate that incorporating artificial intelligence with highly enhanced communication technologies would be important to facilitate real-time decision making in the smart city of the future. Similarly, Revathy et al. [2] This paper has highlighted the importance of predictive analytics in traffic management, where congestion is predicted early enough and can help avert the worst traffic jams. Sharma *et al.* [3] introduced a vision-based vehicle detection system in the urban traffic conditions with the application of deep learning. This research paper indicates the prospect of visual analysis using deep learning as a vehicle detector that can be successfully applied to estimate the density of the traffic and implement intelligent traffic management systems. Dinesh et al. [4] proposed an adaptive traffic signal control system that integrates a vision-based vehicle detection module with dynamic signal timing logic. Their Experimental results demonstrated a reduction in average waiting time and an improvement in traffic flow when compared to fixed-time signal control. The study concluded that adaptive signal control based on real-time traffic observation significantly enhances intersection performance. Dikshit et al. [5] In addition to the optimization of signal timing, several researchers have investigated AI-based traffic optimization using intelligent routing and congestion management techniques. Pillai et al. [6] investigated adaptive traffic flow optimization using artificial intelligence models. The study showed that AI-based traffic control systems outperform traditional methods by continuously adapting to real-time traffic conditions. The authors emphasized the importance of active decision-making mechanisms for managing unpredictable urban traffic. Nirmaladevi *et al.* [7] proposed a smart traffic management system designed to prioritize emergency vehicles using deep learning techniques. Their system detects emergency vehicles through camera-based vision and dynamically modifies signal

phases to ensure faster clearance. The results demonstrated reduced response times without significantly affecting normal traffic flow. Aravind et al. [8] introduced an AI-powered smart traffic signal system incorporating adaptive signal control, emergency vehicle priority, and power optimization. Their experimental results showed improved traffic flow, reduced congestion, and enhanced emergency vehicle movement, highlighting the benefits of integrated intelligent traffic control strategies. Kalaivani et al. [9] proposed an optimized deep learning-based vehicle detection framework for dense urban traffic conditions. Their method achieved improved detection performance under challenges such as occlusions, high traffic density, and varying illumination, demonstrating the importance of accurate vehicle detection for adaptive traffic control. Harsha et al. [10] presented a density-sensitive traffic management approach using image-based congestion estimation. Their system dynamically adjusted traffic control decisions based on real-time congestion levels, showing that vision-based density estimation can effectively improve traffic flow without relying on intrusive physical sensors.

Despite notable advancements in AI-driven traffic management, many existing systems depend on continuous video processing or complex infrastructure, leading to higher computational costs and reduced scalability. Addressing these challenges, the proposed work adopts an image-based vehicle detection framework integrated with adaptive signal timing to deliver a computationally efficient, scalable, and practical solution for intelligent urban traffic control.

III. METHODOLOGY

The aim of the proposed Smart Traffic Management System is to decrease the congestion of traffic in cities through the combination of Artificial Intelligence technologies and computer vision. A trained YOLOv7 model is used to process traffic images on road scenes to identify and classify vehicles, allowing to estimate the density of traffic properly. An adaptive signal control system is used to calculate reasonable green signal times based on the real-time congestion in order to maintain equal traffic flow in all directions. The new signal times are tested in a simulation environment based on Pygame to graphically examine the vehicle movement and signal changes at different traffic volumes.

The algorithm design is based on a sequential pipe comprising of Traffic Image Dataset Preparation, Image Preprocessing, Vehicle Detection, Vehicle Classification and Counting, Traffic Density Estimation, Adaptive Signal Switching, Green Signal Time Calculation, and Traffic Simulation. Figure 1 displays the overall flow of the proposed system.

The stages of the proposed workflow are detailed out under the following subsections:

A. Traffic Image Dataset Preparation

The quality and diversity of the dataset used in training and evaluation largely determine the performance of deep learning models. A traffic image dataset was built in this research based

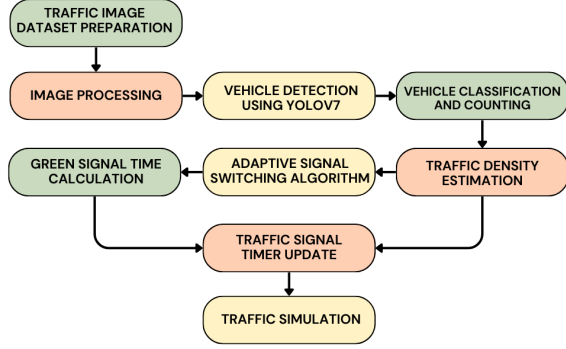


Fig. 1: Workflow Block Diagram of the Proposed Smart Traffic Management System For Urban Congestion

on images gathered from publicly available online sources and open-access traffic image repositories [11]–[13].

The dataset was designed to reflect real-life traffic conditions and included a variety of vehicle types, such as cars, buses, trucks, and motorcycles. Additionally, vehicles used in the simulation setting were obtained from an open-access icon library to visually represent the different vehicle types [14].

To ensure precise localization and classification during the training of the YOLOv7 model, all obtained images were manually labeled. Additional emphasis was placed on incorporating variations in traffic density, viewpoints, and weather conditions to enhance the robustness of the detection system. A well-prepared dataset supports effective feature learning and enhances the ability of the YOLOv7 model to generalize across diverse traffic situations.

B. Image Preprocessing

Before the process of identifying the vehicles, the acquired traffic images were filtered to enable them to fit in the YOLOv7 detection framework and enhance the reliability of the detection. All of the images were downsized to a constant resolution of 416×416 pixels to conform to the requirements of the input of the YOLO architecture, to allow the same extraction of features during training and inference.

These images were then transformed into the representation of a blob with the help of the OpenCV Deep Neural Network (DNN) module that normalizes the values of the pixel intensity and makes it easy to process by the detection model. This preprocessing channel minimizes the differences that occur as a result of varying imaging dimension and lighting scenarios, and maintains the key traffic characteristics, which, in turn, enables the correct vehicle identification and the adaptive signal management.

C. Vehicle Detection using YOLOv7

One of the key aspects of the suggested smart traffic management system is vehicle detection, because precise vehicle recognition is the basis of the traffic density estimation and

adaptive management of the signal. The YOLOv7 (You Only Look Once) object detection model was used in this work because it is capable of offering high-quality and real-time detection at a relatively low computational cost.

YOLOv7 is built on a Convolutional Neural Network (CNN) architecture that processes the entire image in a single forward pass, enabling faster detection than traditional region-based approaches. The architecture comprises three key components: a backbone for hierarchical feature extraction, a neck for multi-scale feature aggregation, and a detection head that predicts bounding boxes, class labels, and confidence scores.

In the suggested system, the OpenCV Deep Neural Network (DNN) module was used to detect vehicles. The input images were translated into the representation of blobs and were sent through the trained network. To reduce weak detections, a confidence threshold was used; then Non-Maximum Suppression (NMS) was used to remove overlapping bounding boxes and keep the best predictions.

The model was able to identify several types of vehicles, such as cars, buses, trucks, motorcycles, and auto-rickshaws. The detections serve as the foundation of vehicle counting and estimation of traffic density, which result in intelligent and adaptive operation of traffic signals.

D. Vehicle Classification and Counting

After vehicle detection, the identified objects were determined as vehicle classification of cars, buses, trucks, motorcycles, and auto-rickshaw, according to the class labels produced by the YOLOv7 model. Correct classification will help in a better understanding of the composition of traffic at the intersection.

Then, the vehicles identified were counted per lane based on region-based assignment, according to the spatial coordinates. The method provides a good quantification of vehicles heading into every signal direction.

The categorized vehicle counts are the crucial input into the estimation of the traffic density, which is the basis of adaptable signal timing and effective traffic flow management in the proposed system.

E. Traffic Density Estimation

The density of traffic was determined as the counts of vehicles in the lanes after classification. These counts are an indication of the level of congestion at the intersection and are used as the foundation of adaptive signal control.

The green signal time needed was calculated with the average clearance time of the vehicle as:

$$G = \left\lceil \frac{\sum (N_i \times T_i)}{L + 1} \right\rceil \quad (1)$$

where N_i is the number of vehicles of type i , T_i is the average time of passage to that type of car, and L is the lanes. The predefined minimum and maximum limits were used to limit the calculated green time to assure stable signal operation. Such density-based estimation allows the system to react dynamically to the changing traffic conditions.

F. Adaptive Signal Switching Algorithm

In the proposed approach, dynamic control of intersection traffic makes use of a density-based, rule-driven adaptive algorithm of signal switching. Signal times are changed based on real-time lane-based vehicle counts that the detection module has received.

The green signal duration needed is calculated as required to accommodate vehicle clearance needs per cycle, in which the larger density lanes are given longer green timing and those of lesser congestion are given shorter timing. The green time is limited to defined limits, so that, $T_g \in [T_{min}, T_{max}]$, ensuring that the operation of the signal is predictable and constant.

The signals move in a cycling sequence to ensure that no approach is left behind and are constantly changing according to the shift of traffic. This control mechanism is an adaptive control that enhances traffic throughput, reduces vehicle waiting time, and allows efficient intersection management.

G. Green Signal Time Calculation

The duration of the green signal is calculated by the density of the traffic estimated according to the formulation in Section III. This is calculated by taking lane-by-lane vehicle counts and mean clearance periods to gain enough crossing time at the intersection.

The computed green interval is dynamically updated for each signal cycle and restricted within predefined minimum and maximum bounds to ensure stable signal operation. This is an adaptive method of time allocation that assists in efficient movement of vehicles and elimination of unnecessary waiting times.

H. Traffic Signal Timer Update

Once the calculation of the required green duration is done, the result is allocated to a specific traffic signal on the controller. The signal timer is based on the principle of a real-time countdown, where the remaining time T_r becomes smaller continuously until it reaches zero, and the signal phases are switched.

The signals are switched sequentially between green, yellow, and red, and then the next signal is activated in a cyclical manner. This dynamic timer action guarantees coordinated movement of signals and also allows efficient implementation of adaptive timing decisions without disrupting the predictable flow of traffic at the crossing.

I. Traffic Simulation

A simulating environment was created to test the capabilities of the proposed adaptive traffic system of controlling at a multi-directional intersection. Cars were dynamically introduced into lanes, and the dynamics of the signals could be observed at different levels of traffic congestion.

There were real-time variations of traffic lights on the adaptive timing, which governed the movement of vehicles in green, yellow, and red lights. The simulated scenario indicated that the adaptive signal switching system is very efficient in controlling traffic flow and improving the overall intersection performance.

IV. RESULTS AND DISCUSSION

The smart traffic management system proposed was tested in a simulated environment to test how well it worked in vehicle detection, estimation of the traffic density, and adaptive signal control. Combining computer vision with dynamic signal timing makes it possible to monitor the conditions at the intersection at all times and provide responsive control over the traffic.

The system uses a traffic image-based vehicle detection module that is based on YOLOv7 and classifies vehicles in traffic images, and then estimates the density of traffic in the lanes. Depending on the congestion levels that are detected, the adaptive signal control system automatically varies signal durations to enhance traffic flows.

The experimental observations show that the suggested strategy can be used to increase the intersection efficiency by distributing the green signal time based on the demand of real-time vehicles. This is a dynamic approach that reduces wastage of time in the waiting lines and enhances better traffic passage than the traditional fixed-time signaling systems.

A. Vehicle Detection Performance

The car detector is a traffic-based vehicle detector, which is based on the YOLOv7 model and is used to detect and label the car in the traffic image. The detector was able to detect various categories of vehicles including cars, buses, trucks, motorcycles, and auto-rickshaws, creating bounding boxes with labels attached to them as shown in Figure 2. In order to further measure the effectiveness of models, standard performance indicators like F1-score and mAP@50 were calculated, as summed in Table I and The outcomes in Table II indicate the existence of the strong-ness and dependability of the suggested YOLOv7-based system..

In testing the effectiveness of the detection model, the standard performance metrics were examined. The findings indicate that the model offers quality detection operation that is appropriate in the estimation of traffic density and adaptive control of signals under the simulation environment.

TABLE I: Performance Evaluation of YOLOv7 Vehicle Detection

Metric	Value
Accuracy	97.8%
Precision	95.0%
Recall	98.0%
F1-Score	96.5%
mAP@50	97.6%

TABLE II: Vehicle Detection Module Summary

Parameter	Observation
Detection Model	YOLOv7
Vehicle Types Detected	Car, Bus, Truck, Motorcycle, Auto-rickshaw
Detection Output	Bounding boxes with class labels
Confidence Threshold	Applied for reliable detection
Multi-object Detection	Supported
System Integration	Enables vehicle counting for density estimation

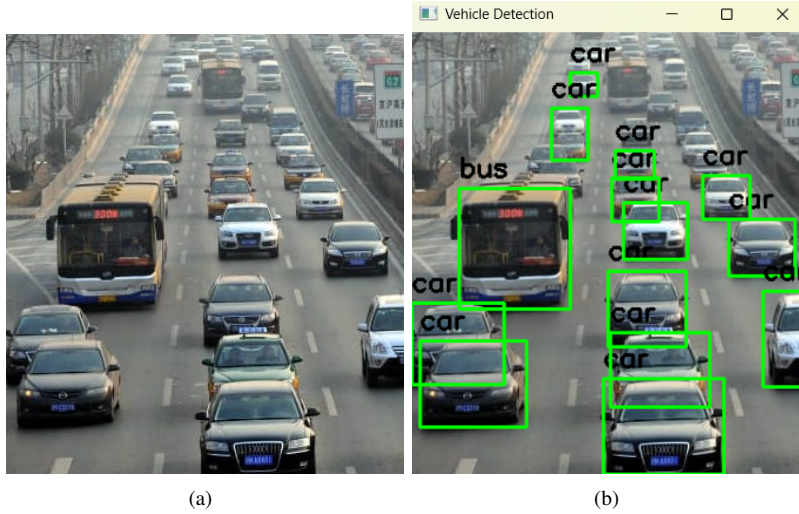


Fig. 2: (a) Input traffic scene (b) Vehicle detection using YOLOv7 with bounding boxes and class labels.

B. Simulation Results

The implementation of the proposed smart traffic management system was checked on a simulation of a four-way intersection developed in Pygame to simulate real-life traffic conditions. Signal timings were dynamically adjusted according to lane-wise vehicle density detected by the YOLOv7 module.

The adaptive control mechanism was used in the allocation of the duration of the green signals to real-time congestion levels instead of fixed time, as is the case with conventional signaling. This plan enhanced vehicle movement by laying emphasis on lanes that are heavily occupied, with minimal wastage of delays at the intersection point.

Figure 3 shows the simulation environment, which emphasizes the movement of vehicles, signal transitions based on the density, and flow of the traffic. The results obtained show that vehicle detection and adaptive signal control increase the overall intersection efficiency in a simulated controlled environment.

C. Comparative Performance Analysis

The comparison was based on literature to analyze the proposed YOLOv7-based vehicle detector framework against the popular object detection models, like SSD, Faster R-CNN, and recent versions of YOLO models. The ability to detect has been found to be strong in these models in traffic monitoring applications.

According to Table III, YOLOv7 also introduces architectural advancement that improves the capacity to extract features and achieve high-detection rates in addition to real-time functionality. These features render it suitable to be applied in adaptive traffic signal systems where speedy and dependable vehicle recognition is paramount.

D. Discussion

Through the experimental findings, it is seen that the suggested smart traffic management system is effective in combining vehicle detection using YOLOv7 and adaptive signal control to enhance the efficiency of the intersection. The system will allow more effective use of the lanes and minimize congestion because the signal durations can be updated dynamically based on the actual traffic density in the area.

In addition, the simulated scenario demonstrates the prospect of integrating computer vision and intelligent control strategies in the current traffic management. The framework proposed gives an area of scalability and practical basis to smart transportation systems designed to increase urban mobility.

V. CONCLUSION

In this paper, an AI-based smart traffic management system has been proposed, which combines the YOLOv7 deep learning model with adaptive signal control to enhance the efficiency of intersections. Traffic images are recognized to identify several types of vehicles by the system, and signal timing can be dynamically changed according to the current traffic density, which allows traffic to be regulated responsively and intelligently.

The suggested framework ensures that the durations of the signals match the current traffic situations and minimizes the unreasonable delays at crossroads. The fact that YOLOv7 is a lightweight architecture and is computationally efficient also contributes to the viability of the application of the system in real-world scenarios of smart cities.

The next generation of work will be aimed at applying the framework to real-world traffic contexts using multi-cameras and implementing more sophisticated learning methods to optimize signal control and increase urban mobility even more.

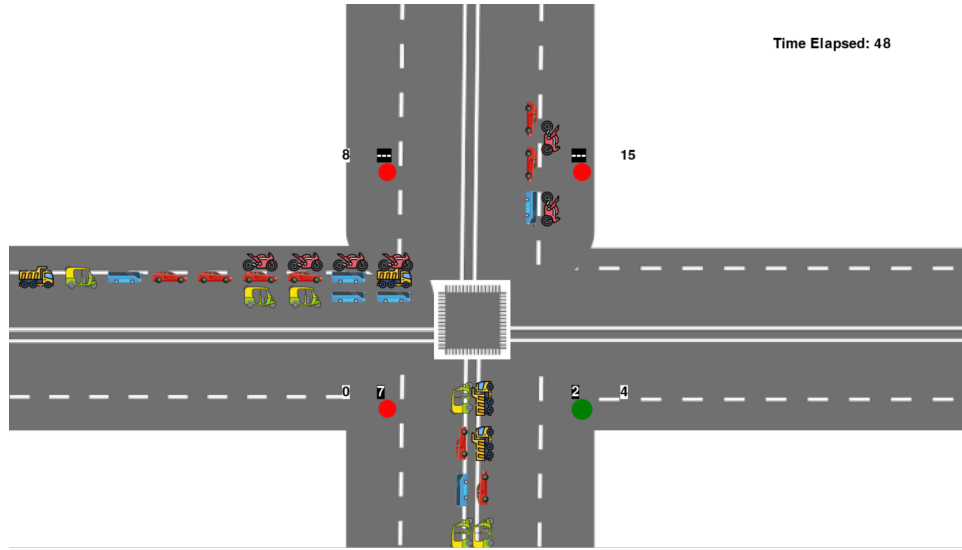


Fig. 3: Traffic simulation illustrating vehicle movement, density estimation, and adaptive signal control at a four-way intersection.

TABLE III: Comparison of Existing Object Detection Models for Traffic Applications

Model	Framework	Accuracy (%)	Precision (%)	Recall (%)
SSD [15]	Single Shot MultiBox Detector	90.5	88.7	89.2
Faster R-CNN [16]	Region Proposal Network	93.6	91.2	92.5
YOLOv4 [17]	CNN-based One-stage Detector	94.5	92.4	93.6
YOLOv5 [18]	Deep Convolutional Backbone	95.3	93.0	95.0
EfficientDet [19]	BiFPN-based Detector	96.0	94.2	94.8
YOLOv7 [20]	Optimized Real-time Detector	97.8	95.0	98.0

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